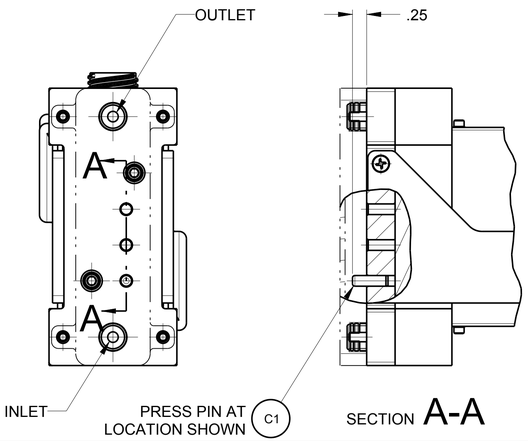
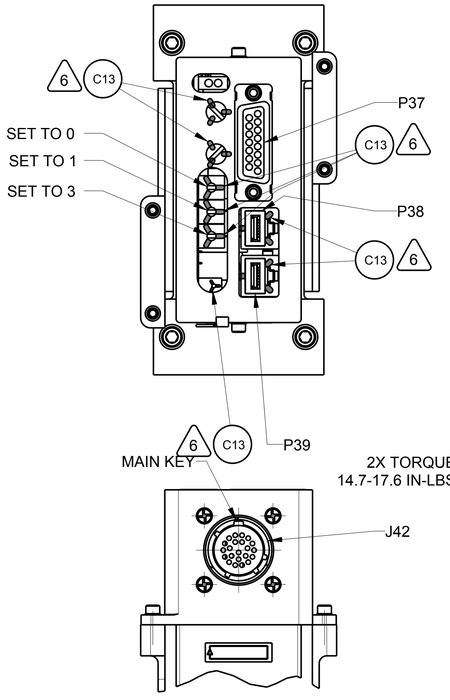
COMMERCIAL PARTS LIST

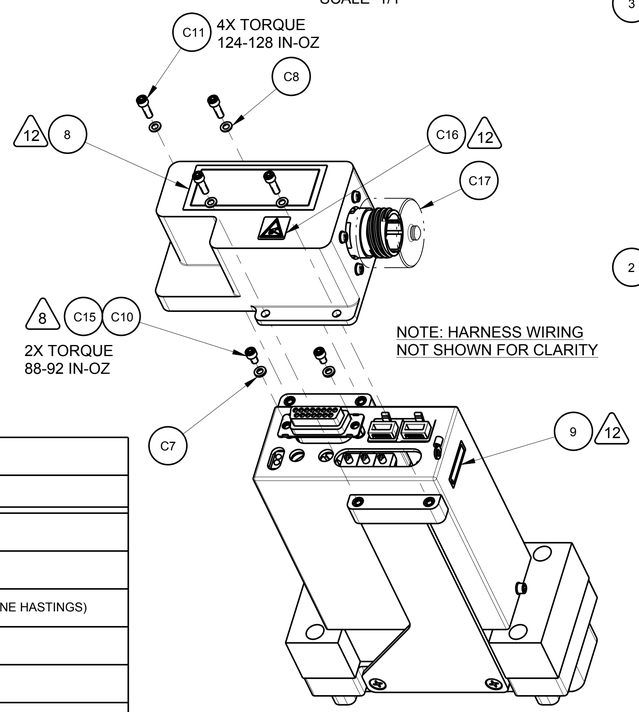
ITEM	QTY	PART NUMBER	DESCRIPTION
C1	1	MS16555-640	HEADLESS STRAIGHT PIN, STAINLESS STEEL .1875 OD X .625 LG, NASM16555
C2	1	M85049/95-14A	NUT PLATE, 3/4 PERIMETER, FLANGE MOUNT RECEPTACLE, SHELL SIZE C
C3	2	M83248/1-109	O-RING, 75 DUROMETER FLUOROCARBON, .299 ID X .103 W
C4	2	M83248/1-012	O-RING, 75 DUROMETER FLUOROCARBON, .364 ID X .070 W
C5	4	NAS1101E04-6	SCREW, MACHINE-FLAT FILLISTER HEAD .112-40 UNJC-3A X .375 LENGTH
C6	4	NAS1102E06-6	.138-32 UNJC-3A X .375 FLAT 100 DEG HEAD MACHINE SCREW
C7	2	NAS620C4	#4 FLAT WASHER REDUCED OUTER DIAMETER
C8	8	NAS620C4L	#4 REDUCED THICKNESS FLAT WASHER REDUCED O.D.
C9	2	NAS620C8L	#8 REDUCED THICKNESS FLAT WASHER REDUCED O.D.
C10	2	NAS1352N04-3	#4-40(.112) UNRC-3A X .188 SOCKET HEAD CAP SCREW
C11	4	NAS1352N04-6	#4-40(.112) UNRC-3A X .375 SOCKET HEAD CAP SCREW
C12	2	NAS1352N08-6	#8-32 (.164) UNRC-3A X .375 SOCKET HEAD CAP SCREW
C13	A/R	3145 GRAY	RTV ADHESIVE/SEALANT, GRAY, DOW CORNING
C14	A/R	MIL-PRF-27617	DUPONT KRYTOX 240AC TYPE III LUBRICANT
C15	A/R	222MS	THREADLOCKER, LOCTITE
C16	1	SDG32104805-009	DECAL, ELECTROSTATIC SYMBOL .5 IN X .5 IN, METAL PHOTO AND ADHESIVE (JSC)
C17	1	660-024M13N	PROTECTIVE COVER, RECEPTACLE, SHELL SIZE 13 (GLENAIR)

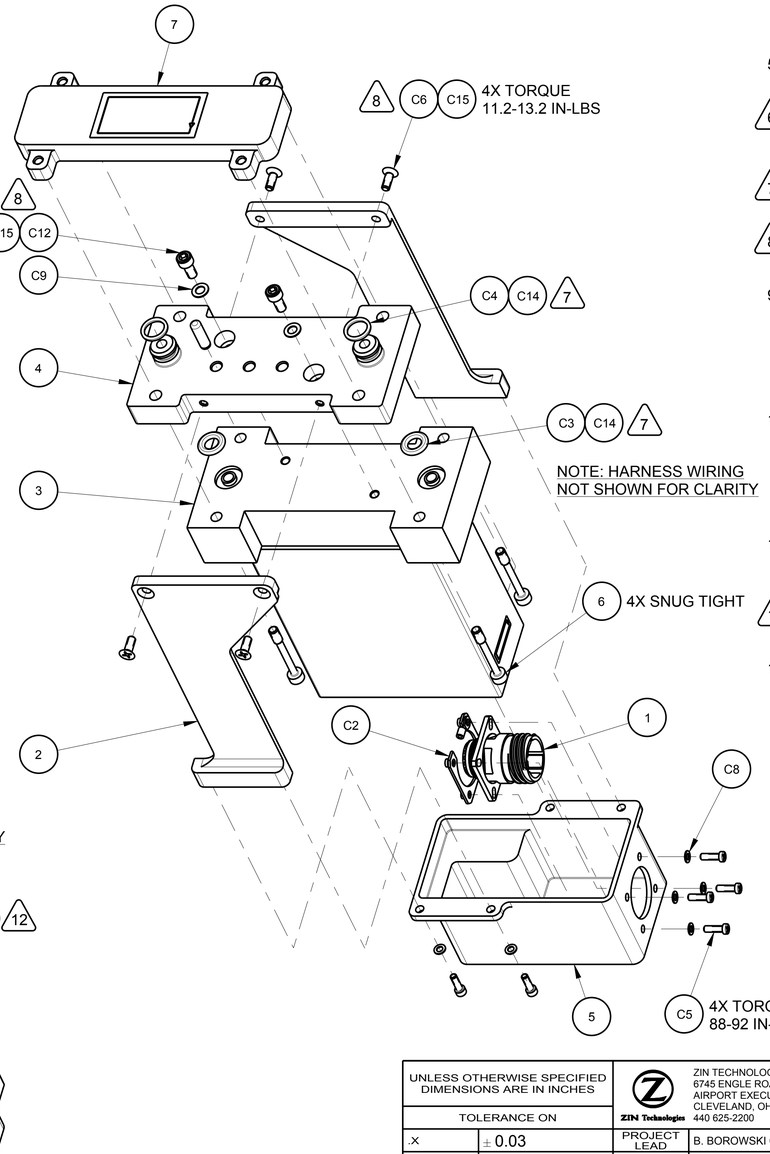


DRAWING LIST

ITEM	QTY	DRAWING NUMBER	DESCRIPTION
1	1	S1101EFA1119	INERT MASS FLOW CONTROLLER HARNESS ASSEMBLY
2	2	S1101MFA1114	MASS FLOW CONTROLLER COVER BRACKET ASSEMBLY
3	1	S1101MFA1140-009	MODIFIED HFC-D-302 0-5 SLM MASS FLOW CONTROLLER (TELEDYNE HASTINGS)
4	1	S1101MFD1112	MASS FLOW CONTROLLER MOUNTING PLATE
5	1	S1101MFD1113	MASS FLOW CONTROLLER COVER
6	4	S1101MFP1116	MASS FLOW CONTROLLER MOUNTING SCREW
7	1	S1101MFA1141	MASS FLOW CONTROLLER INTERFACE COVER ASSEMBLY
8	1	S1101MFT1125-XX	COMBO LABEL
9	1	S1101MFT1125-04	CONNECTOR ID LABEL







14. ENGINEER, OR TECHNICIAN, SHALL FUNCTIONALLY TEST COMPLETED ASSEMBLY PER ACME-TST-030.

15. ENGINEER, OR TECHNICIAN, SHALL PERFORM CONTROL TUNING PROCEDURE ON COMPLETED ASSEMBLY PER ACME-TST-041.

NOTES:

1. CERTIFICATE OF CONFORMANCE REQUIRED FOR ALL COMMERCIAL PARTS.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. CHEMICAL CONVERSION COAT ANY DAMAGED OR MACHINED AREAS ON ALUMINUM PARTS PER MIL-DTL-5541, TYPE I, CLASS 3.
4. CLEAN ALL SURFACES WITH ISOPROPYL ALCOHOL AND LET AIR DRY. WIPE WITH CLEAN COTTON SWABS AND/OR LINT-FREE CLOTH.
5. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
6. STAKE COMPONENTS AT DESIGNATED AREAS SHOWN. STAKING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.1.
7. APPLY O-RING LUBRICANT PER ESTABLISHED PROCEDURES.
8. APPLY THREAD LOCKING COMPOUND PER ESTABLISHED PROCEDURES.
9. TORQUE VALUES NOTED ON THE DRAWING ARE EFFECTIVE TORQUE ONLY. RUNNING TORQUE IS REQUIRED FOR SELF LOCKING DEVICES AND BOLT FRICTION. RUNNING TORQUE MUST BE MEASURED AND ADDED TO THE EFFECTIVE TORQUE. RECORD RUNNING TORQUE AND FINAL TORQUE ON DRAWING.
10. BEFORE STORAGE, PROTECT WITH CONTAMINATION BARRIER. USE CLEAN, NEW, ESD BAG, TOTES OR APPROVED ESD PACKAGING. MARK PART NUMBER, A DASH, AND A SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/SOW, BY TAGGING PER SAE-AS478-35D.
11. REMOVE ALL VENDOR AFFIXED LABELS FROM ITEM 3, MASS FLOW CONTROLLER, AND TEMPORARY LABELS FROM ITEM 1 HARNESS.
12. APPLY LABEL TO CLEAN, DRY SURFACE IN LOCATION SHOWN.
13. SEE SHEET 2 FOR ADDITIONAL INFORMATION.

REV	CHG	ZONE	DESCRIPTION	APP/DATE
E	-		INCORPORATED ACME-ECO-535	AK 11/21/16 LF 11/21/16 AD 11/23/16
D	-		INCORPORATED ACME-ECO-512.	JO 09/28/15 WB 09/28/15 AD 10/02/15 JB 10/14/15
C	-		INCORPORATED ACME-ECO-451.	JO 04/22/15 GF 04/23/15 AD 04/24/15
B	-		INCORPORATED ACME-ECO-297	KR 06/30/14 AD 06/30/14 GF 06/30/14
A	-		INCORPORATED ACME-ECO-116 & ACME-ECO-262.	MA 01/22/14 AD 01/22/14 GF 01/23/14

REVISIONS

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES		
TOLERANCE ON		
.X	± 0.03	PROJECT LEAD: B. BOROWSKI 05/07/12
.XX	± 0.01	QUALITY: D. SEELEY 05/07/12
.XXX	± 0.005	SAFETY: W. BORRELLI 05/03/12
FRACTION	± -	ENGINEER: A. DRAKE 05/04/12
ANGULAR	± 0.50°	DESIGNER: G. FEDEVICH 05/03/12
BREAK EDGES	N/A	REVIEWER: A. ATHERTON 05/04/12

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
440 625-2200

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

ACME CHAMBER INSERT ASSEMBLY, ON-ORBIT CONFIGURATION

S1101MFA0003

INERT MASS FLOW CONTROLLER ASSEMBLY, 5 SLPM

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT APPROVED ECO

APPROVED: S. CORN-CAIN DATE: 10/14/15

DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994

CONTRACT NUMBER: NNC09BA02B
DELIVERY ORDER: SPACE 110
MODEL NAME: S1101MFA1125

PRIOR TO USE CHECK FOR ANY OUTSTANDING ECO'S WITH CONFIGURATION MANAGEMENT OFFICE

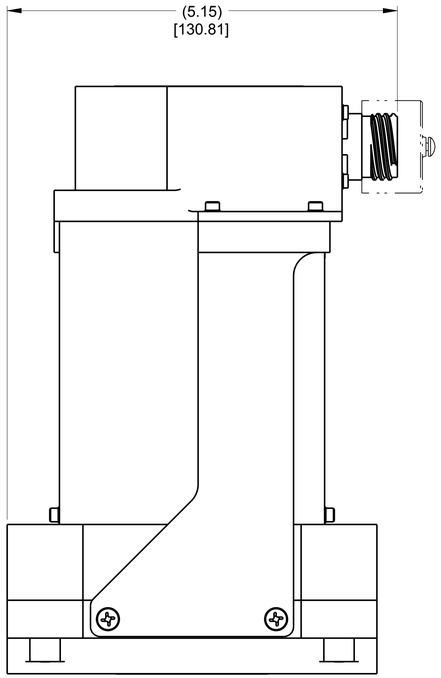
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D	S1101MFA1125	E	ZIN	GRC
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SCALE: 3/4	INITIAL REL DATE: 05/07/2012	SHEET: 1 OF 2	DISCIPLINE: MECHANICAL	

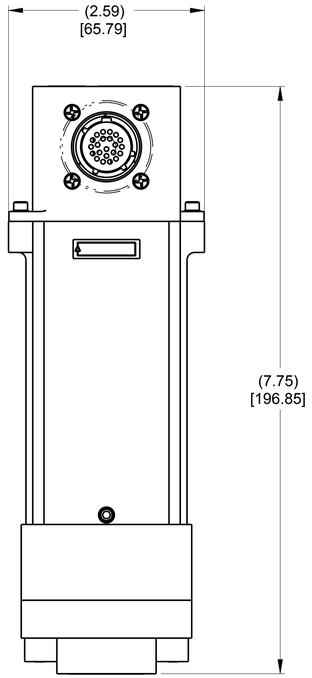
PRO/E VERSION: Wildfire 2.0

SIZE D | DRAWING NUMBER S1101MFA1125 | REV E

NOTES:

1. SEE SHEET 1 FOR ADDITIONAL INFORMATION, GENERAL NOTES AND TOLERANCES.





CHG	ZONE	DESCRIPTION	APP/DATE
REVISIONS			

SIZE	DRAWING NUMBER	REV	BLDG	AREA
D	S1101MFA1125	E	ZIN	GRC
			UNITS: INCH	
SCALE: 1/1	INITIAL REL DATE 05/07/2012	SHEET 2 OF 2	DISCIPLINE MECHANICAL	

PRO/E VERSION: Wildfire 2.0

SIZE D | DRAWING NUMBER S1101MFA1125 | REV E